LESSON QUIZZES

Answers

Letters for multiple choice; model answers for questions 7 and 8 (key words = marks).

MODULE 2 · LESSON 1 · ORGANISMS AND THEIR RELATIONSHIPS

1A 2B 3C 4D 5D 6A

7. A population is individuals of ONE species in the same place at the same time (a school of striped fish). A biological community is a group of interacting populations of DIFFERENT species in the same area (fishes, coral and plants on a reef).

8. The biosphere includes only the parts of Earth that support life: a few kilometers above the surface into the atmosphere and a few kilometers below the ocean surface to the deep-ocean vents. Most of Earth's interior has no life, so the biosphere is a thin layer.

MODULE 2 · LESSON 2 · FLOW OF ENERGY IN AN ECOSYSTEM

1C 2A 3B 4D 5A 6D

7. Dead organisms and wastes would pile up and not be broken down. Nutrients would stay locked in dead matter, so the soil would lose nutrients and producers could not grow well; this would affect every consumer.

8. Producers capture energy from the Sun by photosynthesis. Primary consumers eat producers and get about 10% of their energy; secondary and third-level consumers each get about 10% of the level below. About 90% at each level is used for cellular processes or lost as heat, so available energy decreases going up.

MODULE 2 · LESSON 3 · CYCLING OF MATTER

1A 2C 3B 4D 5B 6A

7. Fossil fuels store carbon from organisms buried long ago (long-term cycle). Combustion releases this carbon as CO_2 much faster than it formed, so the amount of CO_2 in the atmosphere increases.

8. The Sun's energy evaporates ocean water into water vapor. The vapor rises, cools and condenses into cloud droplets. It falls as precipitation (rain) on land, and then percolates through the soil into the groundwater (aquifer).

MODULE 5 · LESSON 1 · BIODIVERSITY

1B 2A 3C 4D 5A 6D

7. Genetic diversity: ladybird beetles of one species with different colors. Species diversity: many different animals at one watering hole. Ecosystem diversity: Arctic ecosystems and tropical rain forests on Earth.

8. Crops come from few species with little genetic diversity, so they can lack disease resistance. Wild relatives (like teosinte for corn) hold useful genes, such as virus resistance, that can be used to improve crops. If they become extinct, those genes are lost forever.

MODULE 5 · LESSON 2 · THREATS TO BIODIVERSITY

1B 2C 3A 4D 5A 6D

7. DDT is not broken down or excreted; it accumulates in tissues. Each organism eats many organisms from the level below, so the toxin becomes more concentrated at each trophic level (biological magnification), reaching the highest level in top carnivores (e.g. 25 ppm).

8. Fragmentation splits a habitat into small parcels. Smaller parcels support fewer species, and populations cannot breed with others across barriers, reducing genetic diversity. More edges also create edge effects (hotter, windier, drier) and more predators and parasites.

MODULE 5 · LESSON 3 · CONSERVING BIODIVERSITY

1B 2C 3A 4D 5B 6A

7. If a few trees are cut from a large forest, new trees can regrow and the forest remains a habitat, so it is renewable. If the entire forest is cleared, there is not enough left to provide habitat; organisms are lost and the forest is nonrenewable.

8. The core zone is protected from human activity, so biodiversity can flourish. The surrounding buffer zones allow sustainable use of natural resources, so local people can still earn a living without destroying the core habitat.

MODULE 1 · LESSON 1 · THE METHODS OF SCIENCE

1A 2B 3C 4D 5A 6C

7. Independent: color of light. Dependent: number of bubbles per minute. Constants: type of plant, water temperature, light intensity, distance from the light. Control: a plant under white (normal) light.

8. Models help when the thing studied is too large, too small, too slow or too dangerous to observe directly. For example, atom models show particles too small to see, and flight simulators let pilots practice dangerous situations safely.